NESA NATIONAL EXAMINATION AND SCHOOL INSPECTION AUTHORITY

SWD - Front-End Design and Development

T2-140

Thursday, 10 July 2025

8:30 AM to 11:30 AM

Names:.....

Index number :.....

TSS NATIONAL EXAMINATIONS, LEVEL 5, 2024-2025

QUESTIONS and ANSWERS BOOKLET

TRADE (s): SOFTWARE DEVELOPMENT (SWD)

SUBJECT: FRONT-END DESIGN AND DEVELOPMENT

DURATION: 3 Hours

Read carefully the instructions on page 3

TOTAL MARKS...../100

FOR EXAMINER'S USE ONLY

QUESTIONS	1	2	3	4	5	6	7	8	9	10	Total
Marks											

QUESTIONS	11	12	13	14	15	16	17	18	19	20	Total
Marks											

QUESTIONS	21	22	23	24	25	26	27	28	29	30	Total
Marks											

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INSTRUCTIONS TO CANDIDATE:

1. This Exam paper is composed of **three sections (A, B and C).** Follow the instructions given below and answer the indicated questions for a total of **100 marks.**

Section **A:**	Fourteen (14) questions which are **Compulsory**	**55 marks**
Section **B:**	Among the five (5) questions, attempt any three (3)	**30 marks**
Section **C:**	Among the two (2) questions, attempt anyone (1)	**15 marks**

2. Fill in your actual **names** and **Index number** on the provided space(Cover).
3. Do not remove or tear any page or part of this Questions and Answers Booklet.
4. Answer in the language in which the Examination is set.
5. In section **A**, answer questions in the provided space. In case of multiple-choice questions, just **circle** the letter corresponding to the correct option.
6. In section **B** and **C**, answer questions in the provided space after section C questions.
7. **Allowed materials:**
   - Blue or Black pen.
   - Geometrical Instruments
   - Silent non-programmable calculators

SECTION A: Attempt all questions **(55 marks)**

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01. Read the following sub-questions and circle the letter corresponding to the right option to each question among the given: **(3Marks)**

i. What is the primary function of the History Panel in Photoshop among the following?

a. To create new layers

b. To apply blending modes

c. To undo multiple actions and monitor changes

d. To adjust color balance

ii. Which tool in Adobe is primarily used for removing small imperfections or blemishes from a photo?

a) Clone Stamp Tool

b) Brush Tool

c) Spot Healing Brush Tool

d) Gradient Tool

02. Read the following sub-questions and circle the letter corresponding to the right option to each question among the given: **(3Marks)**

i. Which of the following best describes user experience (UX)?

a) The process of creating a logo for a website.

b) The overall interaction and satisfaction a user has with a product or service.

c) The visual elements of a website.

d) The hardware used by a user to access a product.

ii. Which of the following is an example of qualitative UX data?

a) Click-through rates on a website.

b) Average time spent on a mobile app.

c) User interview feedback about frustration with navigation.

d) Number of completed sign-ups per day.

03. Based on the description of the Vue.js framework, determine wether the following statements are ***True****(T) or* ***False****(F):* **(3Marks)**

a. Vue.js is primarily used for back-end development and building APIs. .....

b. Vue.js is a JavaScript framework, not a database management system. .....

c. Vue.js helps in designing responsive web pages through CSS, but it is not a CSS framework. .....

04. Match the JavaScript function-related terms with their correct definitions: **(5Marks)**

Terms	**Definitions**	**Answer**
1. Function	**A.** A value that is passed into a function.	1 = ...
2. Parameter	**B.** A block of code designed to perform a specific task.	2 = ...
3. Arrow Function	**C.** An object that represents the eventual completion or failure of an asynchronous operation.	3 = ...
4. Closure	**D.** A function that retains access to its lexical scope.	4 = ...
5. Promise	**E.** A concise way to write functions using the => syntax.	5 = ...

05. Match the **React concepts** in **Column A** with their **descriptions** in **Column B** and write the corresponding answers in the space provided. **(5Marks)**

Column A	Column B	Answer
1. Component	***A.*** A special kind of object that allows developers to write HTML-like syntax in JavaScript.	1 = ...
2. JSX	***B.*** JavaScript features that allow components to interact with lifecycle events or manage	2 = ...
3. Redux	***C.*** A reusable function or class that renders a UI element and its behavior.	3 = ...
4. React Router	***D.*** A tool used for managing global state in an application.	4 = ...
5. Hooks	***E.*** A way to handle navigation and dynamic routing within a React application.	5 = ...

06. Read the sub-questions below and provide appropriate answer in the provided space **(5Marks)**

i. Define AND gate.

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ii. Write the truth table of an AND gate.

07. Complete the paragraph below by using the terms provided in brackets: ***(Panels, Save, Preset, Essentials, Layout, Switch):*** **(3Marks)**

"In Adobe Illustrator, the workspace consists of various tools and **(1)** _____ _____ that help users create and edit graphics efficiently. Illustrator allows users to **(2)** _____ between different workspaces based on their needs. Users can choose from predefined **(3)** _____ workspaces like **(4)** _____ or create their own personalized setup. To keep a customized workspace, users must **(5)** _____ it, ensuring their preferred **(6)** _____ remains available."

08. Determine whether the statements below are ***True****(T)* ***or False****(F)* **(5Marks)**

a. The package.ison file stores all installed dependencies in a Vue project. .....

b. Vue.js is a backend framework. .....

c. The Vue Router package must be installed separately in Vue projects. .....

d. Vuex and Pinia are both used for state management in Vue. .....

e. Vue uses JSX for template rendering by default. .....

09. Read the following sub-questions and circle the letter corresponding to the right option to each question among the given: **(4Marks)**

i. What is the primary purpose of dynamic routes in Next.js?

a) To enforce the primary strict TypeScripting.

b) To create pages based on runtime parameters.

c) To replace API routes to backend logic

d) To disable server-side rendering.

ii. In Next.js, which rendering technique generates pages at build time and serves them as static files?

a) Client-Side Rendering (CSR)

b) Server-Side Rendering (SSR)

c) Static Site Generation (SSG)

d) Incremental Static Regeneration (ISR)

10. Read the following sub-questions and circle the letter corresponding to the right option to each question among the given: **(3Marks)**

i. Which of the following statements, that differentiate linear from non-linear data structures?

a) Linear data structures store elements in a non-sequential order, while non-linear data structures store elements sequentially.

b) Linear data structures store elements in a sequential order, while non-linear data structures store elements in a hierarchical or interconnected manner.

c) Linear data structures can only store numbers, while non-linear data structures can store strings and numbers.

d) Linear data structures are more complex to implement than non-linear data structures.

ii. A database stores 10,000 sorted records. John needs to search for a specific entry efficiently. Which algorithm minimizes the number of comparisons in this scenario?

a) Linear search.

b) Bubble sort.

c) Binary search.

d) Insertion sort.

11. Read the following sub-questions and circle the letter corresponding to the right option to each question among the given: **(3Marks)**

i. Kalisa is designing a 2D platformer game in vue.js where players collect coins and avoid enemies. Which element would display the player's score and remaining lives in the game interface?

a) Splashscreen

b) HUD (Heads-Up Display)

c) Game Environment

d) NPC (Non-Playable Character)

ii. Andy is planning a puzzle game with multiple levels, He wants players to unlock new challenges only after completing previous ones. Which game mechanic enforces this progression?

a) Level gating

b) Dynamic difficulty

c) Cutscenes

d) Randomization

12. What will be the output of the following codes? **(5Marks)**

```
<script type="text/javascript">
var i;
for(i=1;i<=10;i++){
    if(i==3){
        continue;
    }
        if(i==7)
{
            break;
        }
        document.write(i+"<br>");
    }
</script>
```

Answer** or **output

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13. Read the questions below and circle the correct answer to each. **(4Marks)**

i. How can you simulate offline mode in Google Chrome's Developer Tools when testing a Progressive Web Application (PWA)?

a. By disabling the browser's cache

b. By turning off the Wi-Fi on the computer

c. By switching the network status to "Offline" in the "Network" tab

d. By setting the application to "Incognito" mode

ii. In Google Chrome DevTools, which section allows you to inspect the service worker's caching behavior when testing offline functionality?

a. Console

b. Elements

c. Application

d. Sources

14. Read the following sub-questions and circle correct answer to each: **(4Marks)**

i. Which of the following best describes the <a> tag in HTML?

a. It is used to add an image to a web page.

b. It is used to create a hyperlink to another webpage or resource.

c. It is used to display a block of text.

d. It is used to define the structure of a webpage.

ii. What is the benefit of using the <meta> tag in HTML among the following?

a. It defines the main structure of a webpage.

b. It improves the visual appeal of the webpage.
c. It provides metadata such as page description and keywords for search engines.
d. It allows users to interact with forms on the webpage.

Section B: Attempt any three (3) questions **(30 marks)**

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15. Mugabo started selling homemade juice in his neighborhood. Many people liked his juice, but some forgot his business name or mixed up with others. One day, he meets you and you told him that he needs a good logo and a strong brand so that people can remember him. Mugabo wants to understand the difference between a brand and a logo. Help him to **(10Marks)**

Analyze the differences between a **Brand** and a **Logo** by comparing them basing on their features below:

- Definition
- Components
- Purpose
- Longevity
- Emotional connection.

16. Olga is a front-end developer working on a browser-based game using Vue.js at Global Innovation Solution LTD company. She is exploring different styling approaches and is considering using SASS (Syntactically Awesome Style Sheets) for managing the game's visual design. Before deciding, she wants to weigh the pros and cons of using SASS specifically for styling game environments. Basing on Olga scenario discuss **five (5)** advantages and **five (5)** disadvantages of using SASS for styling game environments in a Vue.js project. **(10Marks)**

17. Ganza, a data engineer at a logistic company, is optimizing a system that process thousands of shipment records daily. The records must be sorted stably to maintain the correct order of shipments with the same priority. He needs to evaluate the best sorting algorithm for the task. **(10Marks)**

a. Analyze the performance of different sorting algorithms, such as **Quick Sort**, **Merge Sort**, and **Bubble Sort**, when applied to a large dataset with thousands of records that require stable sorting. In your analysis, consider **brief explanation**, **time complexity** and the **stability** of each algorithm.

b. Recommend Ganza the best sorting algorithm he can use in his project.

18. Mugisha is leading a UX design review for a new app aimed at healthcare professionals. Before finalizing the design mockups, he wants to ensure they truly address the needs of the target users. He asks his team to evaluate the designs against key user-centric criteria. **(10Marks)**

"Evaluate five (5) key factors that should be considered when assessing whether a design mockup effectively meets the needs of the target user group."

19. Read the following questions and circle the letter corresponding to the right option among the given **(10Marks)**

i. What is the primary benefit of using CSS variables compared to traditional hard-coded CSS values in terms of **maintainability**?

a) CSS variables make it easier to modify multiple styles at once by updating a single value.

b) CSS variables provide better performance by reducing the size of the CSS file.

c) CSS variables require less code, making the website load faster.

d) CSS variables make it easier to create animations and transitions.

ii. How do CSS variables contribute to **scalability** in large web development projects compared to traditional hard-coded values?

a) CSS variables allow developers to define reusable values that can be updated across the site, improving consistency and ease of changes.
b) CSS variables reduce the need for CSS rules, making code simpler and more efficient.
c) CSS variables help in reducing the number of lines of code for stylesheets.
d) CSS variables automatically optimize the performance of the website on different screen sizes.

iii. In terms of **maintainability**, which of the following best describes how CSS variables affect future updates to a website's styling?

a) They make it difficult to implement design changes due to their complexity.
b) They allow for centralized management of styles, making future changes quicker and less error-prone.
c) They introduce additional complexity in the styling code, making future updates more challenging.
d) They make it impossible to apply consistent styling to the website across different browsers.

iv. When comparing **CSS variables** to **hard-coded values**, which of the following is true in the context of a **large-scale website**?

a) Hard-coded values allow for faster page loads as there is less code to process.
b) CSS variables offer a more flexible way to handle theme changes, such as switching between light and dark modes, without modifying every individual style.
c) CSS variables require more memory and processing power than hard-coded values, impacting performance.

d) Hard-coded values are better for large websites because they reduce the file size of the CSS.

v. Which of the following statements best describes the **impact of using CSS variables** on **code consistency** compared to traditional hard-coded values?

a) CSS variables often lead to inconsistent styling as they require developers to manage more complex code structures.

b) Hard-coded values are preferable for consistency as they do not rely on variables.

c) CSS variables have no impact on code consistency because they only affect visual elements, not code structure.

d) CSS variables reduce the risk of inconsistent styling by enabling values to be reused across multiple rules, ensuring uniformity throughout the site.

Section C: Attempt only one (1) question **(15 marks)**

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20. You are tasked with creating a web-based application using HTML and JavaScript. The application should allow the user to input a number corresponding to a day of the week and display the corresponding day's name using a switch statement. **(15Marks)**

This is an example of how it should work:

- If the user inputs 1, the output should be "Monday".
- If the user inputs 5, the output should be "Friday".
- If the user enters a number outside the range of 1 to 7, the application should display a message saying, "Please enter a number between 1 and 7."

"You are tasked to write the HTML to accept user input and JavaScript to process the input and display the correct result."

21. Elsie is building a task management app using TypeScript. She wants to define a Task interface with the following attributes: id(number), title(string), and completed(Boolean). **(15Marks)**

Then, she is asked to write a function that takes an existing array of Task objects and a new task, adds the new task to the array, and returns the updated array.

"How would she implement this?"

END

RESERVED SPACE FOR SECTION B AND C ANSWERS

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